FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

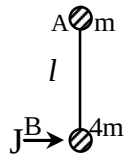
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

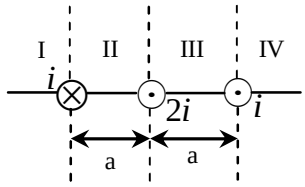
1. An inextensible string connects two masses *m* and 4m and the system rests on a smooth horizontal floor. An impulse *J* is imparted to *B* as shown in figure. The tension in the string in the subsequent motion is



(A) $\frac{J^2}{ml}$ (B) $\frac{J^2}{5ml}$

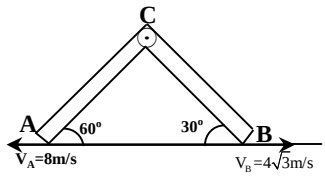
(C) $\frac{4J^2}{5ml}$ (D) $\frac{J^2}{20ml}$

2. Three very long current carrying wires are placed parallel to each other and perpendicular to the plane of paper as shown in the figure. All the wires are equally spaced. The magnetic field will be zero at



(A) Two points in region I
(B) One point in region I and one point in region II
(C) Two points in region III
(D) One point in region I and one point in region III

3. Two rods AC and BC are hinged at the end 'C' and the ends A and B are moving along the surface with velocities $v_A = 8$ m/s and $v_B = 4\sqrt{3}$ m/s respectively as shown. The velocity of hinged point 'C' is

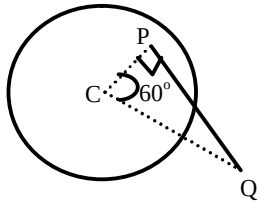


(A) $\sqrt{13}$ m/s (B) $2\sqrt{13}$ m/s

(C) $3\sqrt{13}$ m/s (D) $4\sqrt{13}$ m/s

Space for Rough work

4. A magnetic field, confined in a cylindrical region of radius *R*, is changing at the rate of 4 T/s. A conducting rod *PQ* of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown. The induced emf across the rod will be



(A) $\frac{R^2}{24}(\pi+6)V$ (B) $\frac{R^2\pi}{24}V$

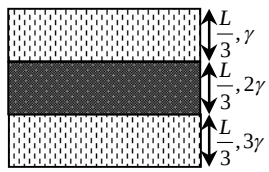
(C) $\frac{R^2\pi}{6}V$ (D) $\frac{R^2}{6}(\pi+6)V$

5. A steel ball of volume $0.02\ \text{m}^3$ is sinking at a speed of 10 m/s in a closed jar filled with a liquid of density $2000\ \text{kg/m}^3$. The momentum of the liquid is

(A) 400 N-s (B) 200 N-s

(C) 100 N-s (D) 300 N-s

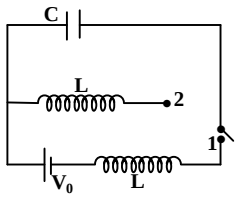
6. Three immiscible liquids are filled in a container as shown. The base area of the container is A and coefficient of cubical expansion of the material of the container is $\frac{3\gamma}{2}$ while the coefficient of cubical expansion of the liquids are shown in the figure. The temperature of the system is increased by ΔT. The volume of the liquid flown out of the container is



(A) $\frac{Al\gamma\Delta T}{3}$ (B) $Al\gamma\Delta T$

(C) $2Al\gamma\Delta T$ (D) $\frac{Al\gamma\Delta T}{2}$

7. At t = 0, switch is closed to position 1. Capacitor is initially uncharged. At $t=\frac{\pi}{2\omega}$, switch is shifted to position 2. Then current as a function of time t is (ω is the angular frequency $=\frac{1}{\sqrt{LC}}$)



(A) $CV_0\omega\cos\ \omega t$ (B) $CV_0\omega(1-\cos\ \omega t)$

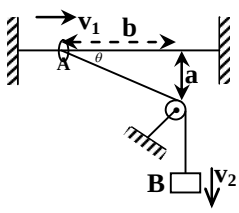
(C) $CV_0\omega(1-\sin\ \omega t)$ (D) $CV_0\omega\sin\ \omega t$

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

8. Ring *A* can slide on a smooth horizontal rod and block *B* can move vertically. Both are attached by an inextensible string. At a certain instant velocity of ring is v_1 and of block is v_2. At that instant ring is at a horizontal distance *b* from pulley as shown. Then



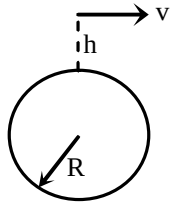
(A) $v_2 = v_1 \cos\theta$

(B) $a_2 = a_1 \cos\theta$, where a_2 *and* a_1 represent the accelerations of block and ring respectively at that instant

(C) $v_1 = v_2 \cos\theta$

(D) Angular velocity of the ring with respect to an observer on the pulley is $\dfrac{v_1 \sin\theta}{\sqrt{a^2 + b^2}}$

9. A particle is taken to a height h from earth's surface and is projected with a velocity *v* as shown. Assume $h << R$, g= 10 m/s^2, *R* = 6400 km. Choose the correct alternative



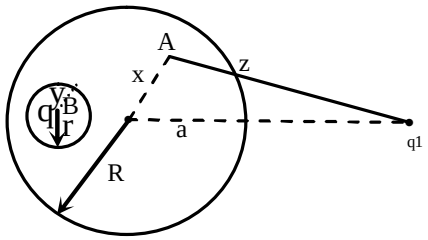
(A) If *v* = 8 km/s, path of the particle will be circular.
(B) If *v* = 10 km/s, path of the particle will be an ellipse.
(C) If *v* = 11.2 km/s, path of the particle will be parabolic.
(D) If *v* = 6 km/s, path of the particle will be an ellipse.

10. In a photoelectric effect experiment, the frequency of incident radiation is increased *n* times $(n > 1)$. Then

(A) Saturation current remains constant
(B) Stopping potential becomes more than n times of initial stopping potential
(C) The difference of the two stopping potentials is equal to $(n-1)\left(V_s + \dfrac{\phi}{e}\right)$, where f is the work function and V_s is the initial stopping potential
(D) The maximum kinetic energy of the photoelectrons increases.

Space for Rough work

11. A cavity is made inside a solid conducting sphere and a charge q is placed inside the cavity at the centre. A charge q_1 is placed outside the sphere as shown in the figure. Point A is inside the sphere and point B is inside the cavity. Then



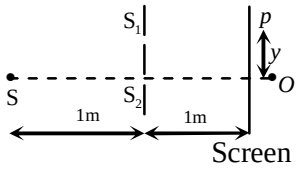
(A) Electric field at point A is zero.

(B) Electric field at point B is $\frac{1}{4\pi\varepsilon_0}\frac{q}{y^2}$

(C) Potential at point A is $\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q}{R}\right]$

(D) Potential at point B is $\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q}{R}+\frac{q}{y}-\frac{q}{r}\right]$

12. In an arrangement shown in the figure, $d << D$, where d is the distance of separation of the slits S_1 and S_2 and D is the distance between the slits and the screen. Source S is emitting monochromatic light of wavelength λ. Then



(A) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{2}}$

(B) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{3}}$

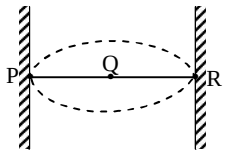
(C) For this value of d, the fringe width is $\sqrt{2\lambda}$

(D) For this value of d, the fringe width is $\sqrt{\frac{\lambda}{2}}$

13. An escalator is moving down with constant speed. You are moving on it such that you remain at rest with respect to ground. Choose correct statements from ground frame:
(A) Work done by you is zero
(B) Work done by escalator on you is zero
(C) Work done by gravity on you is zero
(D) Work done by escalator on you is negative

Space for Rough work

14. A string *PQR* is fixed between two rigid supports separated by distance *L*. The string is vibrating in its fundamental mode as shown in the figure. Choose the correct alternative(s). (Point *Q* is the mid-point of the string)



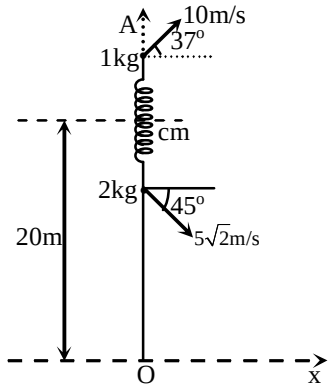
(A) Point *Q* on the string will never have potential energy.
(B) Energy of the portion *PQ* is constant.
(C) Point *Q* on the string will have maximum potential energy when its kinetic energy is zero.
(D) Energy of the string *PQR* is constant but transfer of energy takes place between portions *PQ* and *QR*.

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

A system consists of two small spheres of mass 1kg and 2kg interconnected by a weightless spring. At t = 0, the spheres are set in motion with initial velocities as shown in figure. Position vector of centre of mass at t = 0 is $20\,\hat{j}$. Initially spring is in its natural length. System moves in earth's gravitational field.



15. The momentum of the system at t=2sec in kg m/s is $(g = 10 m/s^2)$

(A) $\left(12\hat{i} - 32\hat{j}\right)$ (B) $\left(16\hat{i} - 64\hat{j}\right)$

(C) $\left(18\hat{i} - 64\hat{j}\right)$ (D) $\left(12\hat{i} - 18\hat{j}\right)$

Space for Rough work

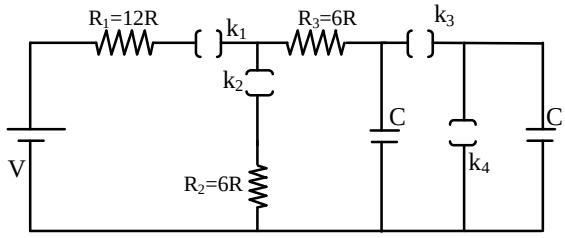
16. The position vector of centre of mass of the system at *t*=1*s* is

(A) $4\hat{i}+\frac{24}{25}\hat{j}$ (B) $6\hat{i}+\frac{39}{2}\hat{j}$

(C) $4\hat{i}+\frac{29}{3}\hat{j}$ (D) $6\hat{i}+\frac{41}{3}\hat{j}$

Paragraph for Question Nos. 17 and 18

The ability of a capacitor to store potential energy is the basis of defibrillator devices, which are used in emergency by medical team to stop the fibrillation of heart attack victims. In a portable defibrillator device, a battery charges a capacitor to a high potential difference, storing a large amount of energy in less than a minute. The battery maintains only a modest potential difference; an electronic circuit repeatedly uses that potential difference for greatly increasing the potential difference of the capacitor. When a control switch is closed, the capacitor lends a portion of its stored energy from paddle to paddle through the victim. The figure shows an arrangement, to understand the working of a defibrillator.



17. If k_1, k_2 *and* k_3 are closed (keeping k_4 open) at $t=0$, then current in the resistor R_2 at $t=t$ is given by

(A) $\frac{V}{15R}e^{-\frac{t}{10RC}}$ (B) $\frac{V}{11R}e^{-\frac{t}{20RC}}$

(C) $\frac{V}{90R}\left(1+5e^{-\frac{t}{20RC}}\right)$ (D) $\frac{V}{90R}\left(5-2e^{-\frac{t}{20RC}}\right)$

18. Long time after closing k_1, k_2 and k_3, if k_1, k_3 are opened and k_4 is closed, then heat produced in R_2 will be

(A) $\frac{1}{36}CV^2$ (B) $\frac{1}{9}CV^2$

(C) $\frac{1}{4}CV^2$ (D) $\frac{1}{6}CV^2$

Space for Rough work

Chemistry

PART – II

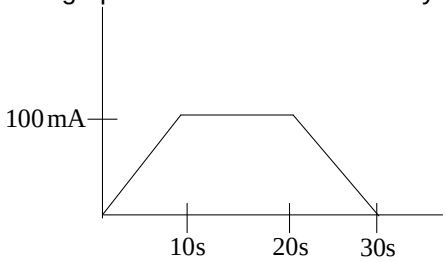
SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Of the following, which will not show chromyl chloride test?
(I) Rubidium chloride
(II) Benzene diazonium chloride
(III) K_2PtCl_6
(IV) Mercurous chloride
(A) II, III and IV (B) III and IV
(C) IV only (D) II and III

20. Marbles of diameter 2cm are to be placed either inside or upon an equilateral triangle (edge length 4cm) drawn on a floor. The maximum number of marbles that can be accommodated is/are:
(A) 1 (B) 2
(C) 3 (D) 6

21. In a Cu voltameter, mass deposited in 30 seconds in 100 grams. Carefully analyse the current-time graph shown below and identify the incorrect statement:



(A) Electrochemical equivalent for Cu is 50
(B) A constant current of 66.66 mA would also discharge the same amount in the same time.
(C) 33.33 grams got discharged in 10 seconds.
(D) 50 grams got discharged in 15 seconds.

Space for Rough work

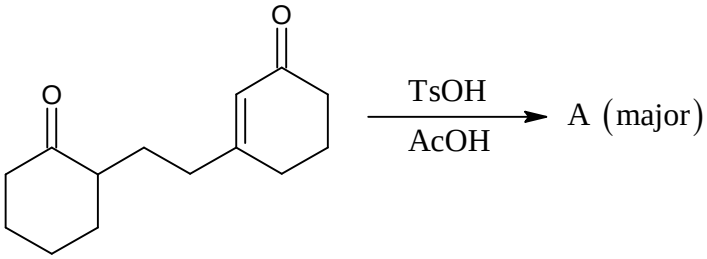
22. Which of the following statements is correct about the conduction of electricity in pure crystal of silicon at room temperature?
(A) The conduction is due to the holes formed from the release of electrons
(B) The conduction is due to the movement of both the electrons released and holes formed
(C) The conduction is only due to the electrons present in fully occupied lowest energy states
(D) The conduction is due to only some electrons capable of leaving the bonds at room temperature

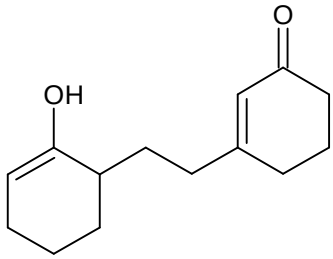
23. The vapour pressures of benzene, toluene and a xylene are 75 Torr, 22 Torr and 10 Torr at $20°C$. Which of the following is not a possible value of the vapour pressure of an equimolar binary/ ternary solution of these at $20°C$? Take all solutions as ideal in nature.
(A) $48\frac{1}{2}$ (B) 16
(C) $35\frac{2}{3}$ (D) $53\frac{1}{2}$

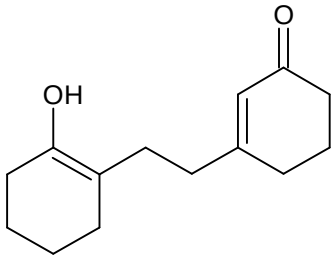
24. White phosphorus when dissolves in boiling $NaOH$ solution in an inert atmosphere, moles of salt of an inorganic salt formed if 24.8 gm of white phosphorus is taken.
(A) 0.2 (B) 0.3
(C) 0.4 (D) 0.6

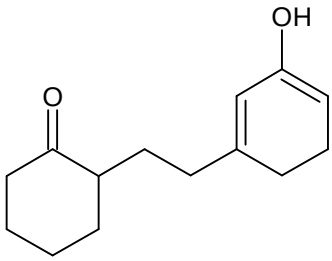
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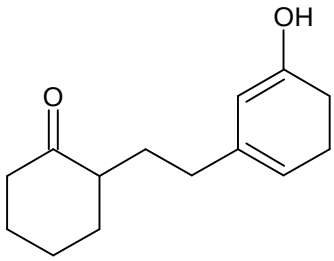
25. Which enol form is involved in this intramolecular aldol?



(A) 

(B) 

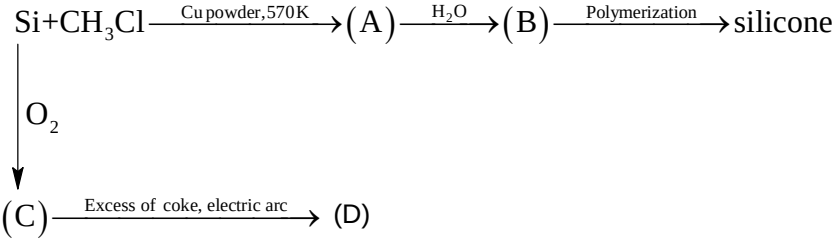
(C) 

(D) 

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. Identify A, B, C and D in the following reaction sequence.



(A) The compound A is $(CH_3)_2SiCl_2$
(B) The compound C and D are SiO_2 and SiC respectively
(C) The compound B is $(CH_3)_2 Si(OH)_2$
(D) The compound B is $(CH_3)_2 Si = 0$

Space for Rough work

27. When colourless crystals of KI are added to a beaker containing hydrogen peroxide, a clear solution(S) is formed along with a grey-violet precipitate (P). Identify the correct statement:
(A) S reacts with ammonium salts to evolves NH_3.
(B) S conducts Saponification of tristearin to 'hard soap'
(C) P dissolves in CCl_4
(D) P does not dissolve in an aqueous solution of KI

28. The volume of which of the following unit cells will be independent of α, β, γ (interfacial angles):
(A) Orthorhombic (B) Monoclinic
(C) Hexagonal (D) Rhombohedral

29. A complex is represented as $CoCl_3.xNH_3$. It's 0.1 molal solutions in water shows $\Delta T_f = 0.558°$. If the cryoscopic and Ebullioscopic constants for H_2O are 1.86 and 0.52 respectively, identify the correct versions assuming 100% ionisation of the complex:
(A) Hexaammine cobalt (III) chloride is the IUPAC name for the complex
(B) The above solution will boil between 100°C to 101°C
(C) The central metal ion is d^2sp^3 hybridised
(D) The coordination sphere is incapable of exhibiting geometrical isomerism

30. Which of the following is/are oxidising agent?
(A) H_2S (B) H_2SO_5
(C) $H_2S_2O_8$ (D) H_3PO_2

31. Identify the correct statements regarding $\left[Cr(en)_2Cl_2\right]^+$ *and* $\left[Co(C_2O_4)_2(NH_3)_2\right]^-$ complex ions:
(A) Both are chelate complexes
(B) Both have octahedral geometry
(C) Both can have three – three geometrical isomers
(D) Both are organometallic compound

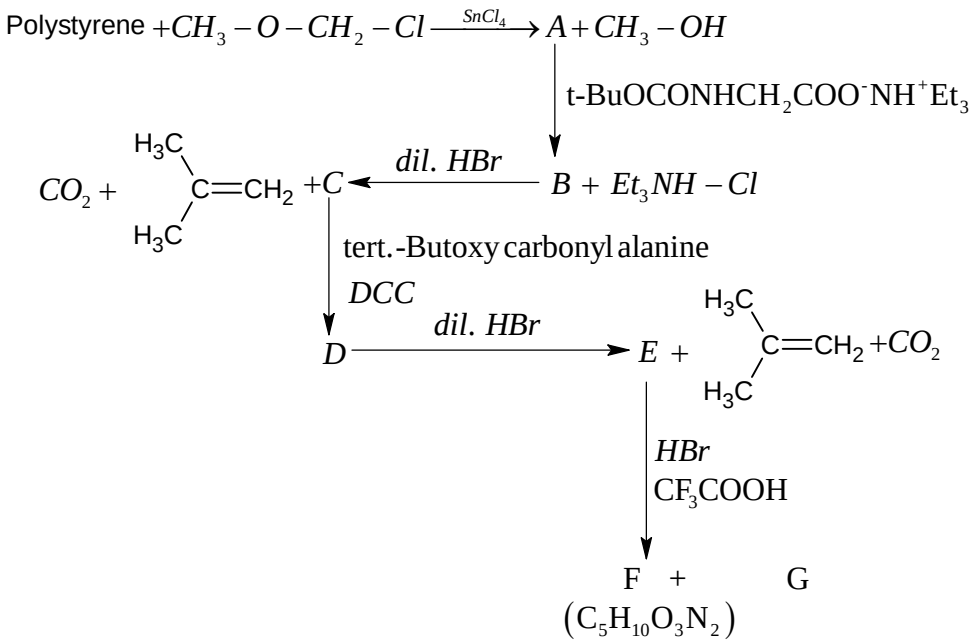
32. The correct statements about some of the ions of 3d-series in aqueous solution
(A) Cr^{2+} is better reducing agent than Fe^{3+}
(B) Co^{3+} is better oxidising agent than Fe^{3+}.
(C) When Fe^{3+} is used as oxidising agent, it attains $3d^7$ electronic configuration
(D) Cu is the only metal in 3d series, which is not having ability to displace hydrogen as H_2 from acidic solution (consider thermodynamic aspects).

Space for Rough work

Paragraph Type (One Option Correct)

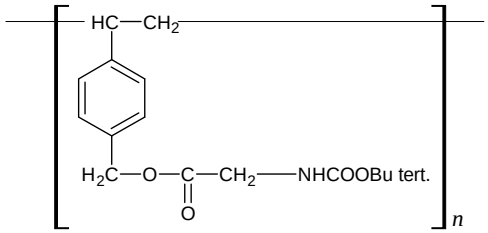
This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 33 and 34

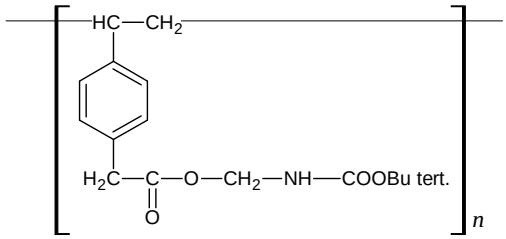


33. The compound 'B':

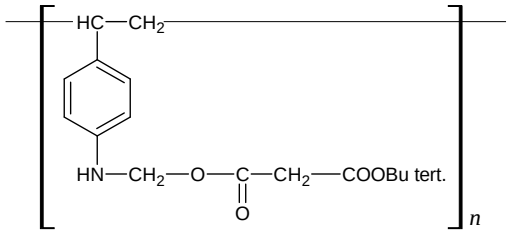
(A)



(B)



(C)



(D) None of these

Space for Rough work

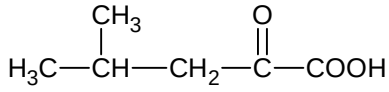
34. The F will be:
(A) Alanyl glycine
(B) glycyl alanine
(C) Alanine
(D) Lecinyl glycine

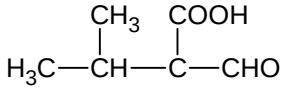
Paragraph for Question Nos. 35 and 36

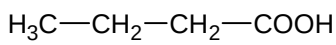
Ethyl isovalerate + ethyloxalate $\xrightarrow{NaOC_2H_5}$ $A\left(C_{11}H_{18}O_5\right)$

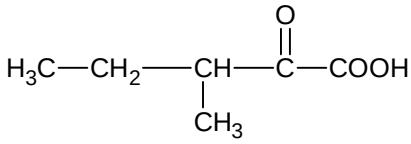
$A+10\%\,H_2SO_4 \xrightarrow{\text{boil}} B\left(C_6H_{10}O_3\right)+CO_2+C_2H_5OH$

35. The structure of 'B' will be:

(A) 

(B) 

(C) 

(D) 

36. $B+NH_3+H_2 \xrightarrow{Pd,\,\Delta} C$, natural occuring amino acid:
'C' will be:
(A) leucine
(B) isoleucine
(C) alanine
(D) glycine

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Suppose f and g are functions having second derivatives f" and g" everywhere, if f(x) g(x) = 1 for all x and f ' (x) and g' are never zero, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ equals

(A) $\frac{f'(x)}{f(x)}$ (B) $2\frac{f'(x)}{f(x)}$

(C) $-\frac{2g'(x)}{g(x)}$ (D) $-\frac{2f'(x)}{f(x)}$

38. Let a real valued differentiable function f(x) be strictly monotonic on [a, b], then

(A) $\int_a^b \left[(f(x))^2 - (f(b))^2\right]dx = \int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx$

(B) $\int_a^b \left[(f(x))^2 - (f(a))^2\right]dx = \int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx$

(C) $\int_a^b [f(x) - f(a)]^2\,dx = \int_{f(a)}^{f(b)} x\left(b - f^{-1}(x)\right)^2 dx$

(D) None of these

39. Let A be the point of intersection of the curve $y = \log_{\frac{1}{3}}\left(x - \frac{1}{3}\right) + \log_3\sqrt{9x^2 - 6x + 1}$ and the circle $x^2 + y^2 = 17$. B is also a point on the curve but lies inside the given circle such that its abscissa is an integer, then max {AB} is

(A) 2 units (B) 3 units

(C) 4 units (D) None of these

Space for Rough work

40. If S_n denotes the sum of first n terms of an AP and $a_n = \frac{S_{3n}}{S_{2n} - S_n}$, then $\underset{n\to\infty}{Lt} \sum_{r=1}^{2n} \frac{a_r}{n}$ is equal to

(A) 2 (B) 4
(C) 6 (D) 8

41. The possible value of a for which $e^{2x} - (a-2)e^x + a < 0$ holds for at least one positive x is
(A) 2 (B) 3
(C) 4 (D) 7

42. Solution of the differential equation $\frac{dy}{dx} = \frac{3x^2y^4 + 2xy}{x^2 - 2x^3y^3}$ is

(A) $x^2y^3 + \frac{x}{y^2} = c$ (B) $x^3y^2 + \frac{x^2}{y} = c$

(C) $\frac{x^2}{y^3} + xy^2 = c$ (D) $x^3y^2 - \frac{x^2}{y} = c$

43. Angle between $\bar{a}$ & $\bar{b}$ is $\frac{\pi}{4}$; $|\bar{a}| = 2\sqrt{2}$, $|\bar{b}| = 3$. Length of longer diagonal of the parallelogram having adjacent sides represented by $5\bar{a} + 2\bar{b}$ and $\bar{a} - 3\bar{b}$ is
(A) $\sqrt{593}$ (B) $\sqrt{26}$
(C) 15 (D) none of these

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. $\int_{-1}^{1} \left(e^{x^3} + e^{-x^3}\right) dx$ is less than

(A) 2 (B) $2e + \frac{2}{e}$

(C) $e + \frac{1}{e} + 2$ (D) 2e

Space for Rough work

45. Let $f(x)=e^{-\frac{1}{x^2}}+\int_0^{\frac{\pi x}{2}}\sqrt{1+\sin t}\,dt \forall x\in(0,\infty)$ then-

(A) f' exists and is continuous $\forall x\in(0,\infty)$

(B) f'' exists $\forall x\in(0,\infty)$

(C) f' is bounded

(D) There exists $\alpha>0$ such that $|f(x)|>|f'(x)|\forall x\in(\alpha,\infty)$

46. Let x,y,z are positive real numbers and ℓ_1 is the least value of $2x^4+2y^4+4z^4-8xyz$ and ℓ_2 is the least value of $x^4y+xy^4+\frac{4}{x^2y^3}+\frac{1}{x^3y^2}+8$, then-

(A) $\ell_1=-1$ (B) $\ell_2>-1$

(C) $\ell_2=10$ (D) $\ell_2>10$

47. Consider the equation in x, $8x^4-16x^3+16x^2-8x+a=0$, then sum of all the non-real roots of equation can be $(a\in R)$

(A) 1 (B) 2

(C) $\frac{1}{2}$ (D) None of these

48. For the equation $\frac{40}{x-1}-\frac{160}{x-4}-\frac{200}{x-5}+\frac{320}{x-8}=6x^2-27x$

(A) Number of real solution of above equation is 3

(B) If E denotes the product of non-zero or complex roots of equation, then sum of divisors of E is 2904

(C) It S denotes the set of all real roots of equation then, sum of elements of S taken two at a time is 81

(D) If $\alpha_1,\alpha_2\in R$ be two roots of equation such that $\log_{\alpha_2}(2\alpha_1)$ is defined it must be 1.

Space for Rough work

49. Let a, b, c be the positive integers such that a<b<c. If the two curves $y = |x-a| + |x-b| + |x-c|$ and $2x + y = 2003$ have exactly one point in common, then-
(A) Least possible value of c is 1002
(B) Greatest possible value of b is 1001
(C) Least possible value of b is 1002
(D) Greatest possible of a is 1000

50. Consider the equation tan ($5\pi \cos\theta$) = cot ($5\pi \sin\theta$) for $\theta \in (0, 2\pi)$, then
(A) number of real solution is 28 (B) number of real solution is 14
(C) $\sin\theta + \cos\theta = \frac{2n+1}{10}$ (D) $\sin\theta - \cos\theta = \frac{2n+1}{10}$

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

Locus of the centre of a circle touching the circle $x^2 + y^2 - 4y - 2x = \frac{7}{4}$ internally and tangents on which from (1, 2) is making an angle 60° with each other is director circle (*C*) of a variable ellipse *E*. On this basis given answers to the following questions

51. The equation of the director circle (*C*) is
(A) $(x-2)^2 + (y-1)^2 = \sqrt{3}$
(B) $(x-1)^2 + (y-2)^2 = 3$
(C) $(x+1)^2 + (y-2)^2 = \sqrt{3}$
(D) $(x-1)^2 + (y+2)^2 = 3$

Space for Rough work

52. Locus of focus of variable ellipse (*E*) when major axis is parallel to *x*–axis, is

(A) $\left\{(x,\ y)/\sqrt{\frac{3}{2}} < |\ x-1\ | < \sqrt{3} \text{ and } y = 2\right\}$

(B) $\left\{(x,\ y)/\ 0 < |\ x\ | < \sqrt{3} \text{ and } y = 2\right\}$

(C) $\left\{(x,\ y)/\ 0 < |\ x+1\ | < \sqrt{3} \text{ and } y = 2\right\}$

(D) $\left\{(x,\ y)/\ 0 < |\ x-1\ | < \sqrt{3} \text{ and } y = 2\right\}$

Paragraph for Question Nos. 53 and 54

Let S = {1, 2, 3,…..n} be a set and two numbers x and y are drawn with replacement from S. Let P_k be the probability that $x^2 - y^2$ is divisible by k & [x] denotes greatest integer less than or equal to x.

53. The value of P_2 is equal to

(A) $\frac{1}{2}$

(B) $\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

(C) $1-2.\frac{\left[\frac{n}{2}\right]}{n}+2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

(D) $1-\frac{2-\left[\frac{n}{2}\right]}{n}-2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

54. The value of P_3 is equal to

(A) $\frac{\left[\frac{n}{3}\right]}{n}$

(B) $\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

(C) $1+\frac{\left[\frac{n}{3}\right]}{n}+\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

(D) $1-2.\frac{\left[\frac{n}{3}\right]}{n}+\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

Space for Rough work